

CHANDIGARH UNIVERSITY

Institute of Distance & Online Learning (IDOL)



Quizminia

AI-Adaptive Learning Platform with Real-Time Difficulty Adjustment

Live Project URL: <https://quizminia.examprap.click/>

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THE ONE-SIZE-FITS-ALL DEFICIT

Traditional online quiz environments treat all learners identically. They administer standard, linear questionnaires regardless of individual student cognitive boundaries.

Standard testing frameworks create cognitive overload for struggling students and trigger boredom for high-achievers.

× Feedback Deficit

Static explanations that ignore student error vectors and fail to tutor the student dynamically.

Secure Portal Entry



QuizMinia

AI-Adaptive Learning Platform

Email Address

teacher@quizminia.com

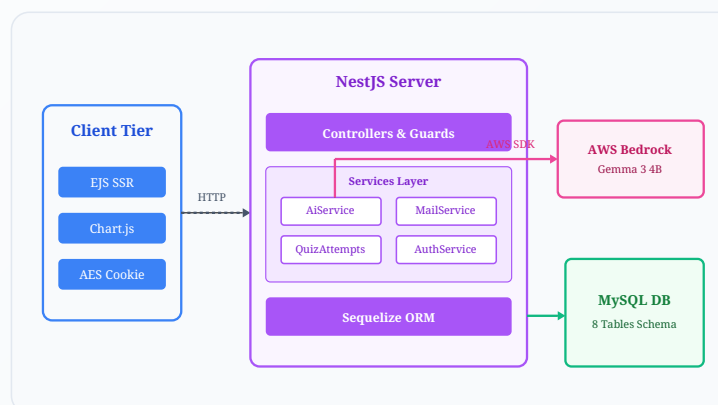
Password

Sign In

THE QUIZMINIA SOLUTION

Quizminia bridges the gap between evaluation and personalization. By incorporating an AI difficulty feedback controller and clean student analytics, the platform maintains a continuous flow zone for the student.

- ✦ **Adaptive Question Routing:** Adjusts difficulty dynamically (0.0 to 1.0) after every answer.
"Keeping students challenged without causing frustration or cognitive fatigue."
- ✦ **High System Availability:** Implements a local rule-based safety fallback engine to handle Bedrock API offline states.
- ✦ **Intelligent Analytics:** Tracks student strengths, weaknesses, and average difficulty baselines.
- ✦ **Robust Security Infrastructure:** Employs PBKDF2 hashing, AES-256 session cookie encryption, and roles guards.



FRAMEWORK TECH STACK

Quizminia leverages a highly modular and secure framework combination to ensure enterprise stability:

- ✦ **NestJS 11 & TypeScript:** Strict MVC controller-service separation, enabling testable and maintainable core code.
- ✦ **Sequelize ORM & MySQL 8:** Transaction-safe schema definitions with robust relational query parameters.
- ✦ **AWS Bedrock & Gemma 3:** Cloud-based serverless execution of the 4B parameters LLM for instant response scoring.

RELATIONAL ENTITY LAYOUT

The MySQL schema consists of 8 tables structured with strict foreign keys and automated database level cascades:

- ✦ **Users & Roles:** Separates students, teachers, and admins, storing PBKDF2 hashes and performance baseline profiles.
- ✦ **Quizzes & Questions:** Base entities containing subject categories, base difficulty scores, and questions.
- ✦ **Attempts & Answers:** Tracks live attempt indicators, accuracy streaks, response feedback, and custom AI insights.

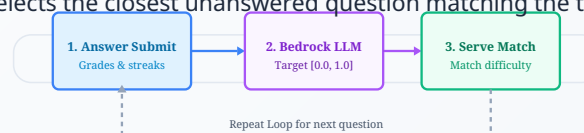
CORE ENTITIES SCHEMA

```
users: [id, email, password, roleId, firstName, performanceProfile]
quizzes: [id, title, subject, difficulty, timeLimit, createdBy]
questions: [id, quizId, text, options, correctAnswer, difficultyScore]
quiz_attempts: [id, userId, quizId, currentDifficulty, score, status]
quiz_attempt_answers: [id, attemptId, questionId, selectedOption, isCorrect]
notifications: [id, userId, title, message, type, isRead]
```

REAL-TIME LOOP CALIBRATION

Difficulty sits on a float scale from 0.0 (easiest) to 1.0 (hardest). The closed-loop controller behaves as follows:

- ✦ **Answer Submission:** Student answer triggers streak calculations.
- ✦ **AI Inference:** The AiService packages history, streaks, and scores, calling Gemma 3 on AWS Bedrock for a new difficulty target.
- ✦ **Adaptive Matching:** The engine selects the closest unanswered question matching the target difficulty from the database pool.



HIGH AVAILABILITY SAFEGUARDS

External LLM APIs face latencies, credential expirations, or rate limits. To guarantee uninterrupted test attempts, Quizminia incorporates a local deterministic fallback engine.

If Bedrock fails, the exception is caught, and difficulty scales rule-based according to streak dynamics.

```
// Rule-based Fallback Step Math
if (isCorrect) {
  if (correctStreak >= 3) diff += 0.20;
  else if (correctStreak == 2) diff += 0.10;
  else diff += 0.05;
} else {
  if (wrongStreak >= 3) diff -= 0.20;
  else if (wrongStreak == 2) diff -= 0.10;
  else diff -= 0.05;
}
diff = Math.max(0, Math.min(1, diff));
```

Guarantees zero-downtime test completion on poor network connections.

PERFORMANCE TRACKING & HISTORY

Students access a dedicated page containing historical quiz scores, active attempts, and weakness metrics:

- ✦ **Real-Time Graphing:** Chart.js charts plot student scores and difficulty parameters across quizzes.
- ✦ **Adaptive Profiles:** System maps strengths/weaknesses and details current baseline levels.
- ✦ **Notifications Inbox:** Receives alerts when new tests are configured.

Student Dashboard View

John
Logout

Dashboard
Available Quizzes
My History
Notifications

Welcome Back, John! 🙌

Your learning profile is initialized. Take a quiz to let the AI adapt to your skills!

COMPLETED QUIZZES
6

AVERAGE SCORE
49.63%

SUCCESS RATE
33%

AI TIER LEVEL
Easy

Recent Activity

Quiz	Score	Outcome	Completed
Algebra Basics Mathematics	100.0%	Passed	5/23/2026
Science Fundamentals Quiz Science	40.0%	Failed	5/23/2026

Learning Profile

STRENGTHS

exponents
powers
arithmetic
variables
addition
subtraction
simplification
algebra
expressions
substitution
equations
linear
solving

MANAGEMENT & ASSESSMENT CONTROLS

Educators have specialized controls to design and grade assessments efficiently:

- ✦ **Quiz CRUD Builder:** Define subjects, difficulty, time constraints, and questions.
- ✦ **Student Insights:** Track average classroom scores, pass rates, and completion times.
- ✦ **AI Generation:** Automated question suggestions matching difficulty metrics.

Teacher Dashboard View

JD

Jane Doe
Mathematics

Sign Out

Dashboard

Quizzes & Questions

Student Results

3

Total Quizzes

2

Published Quizzes

1

Student Attempts

33%

Avg Pass Rate

Quiz Analytics Breakdown

Quiz Title	Subject	Adaptive	Published	Attempts	Avg Score	Pass Rate
Dynamic SA Quiz	Math	AI Adaptive	Draft	0	0%	0%
Mathematics Fundamentals	Mathematics	AI Adaptive	Published	1	77.78%	100%
Science: Forces & Motion	Science	AI Adaptive	Published	0	0%	0%

SYSTEM PROTECTION LAYERS

Student data privacy is enforced backend-side. The system implements cryptography models rather than relying on basic text storage:

- ♦ **PBKDF2 Password Hashing:** Avoids database exposures by executing 1000 iterations of SHA-512 key stretching with unique salts.
- ♦ **AES-256 Cookie Encryption:** Session identifiers are encrypted using AES-256-CBC with an initialization vector, decrypted server-side in session middleware.
- ♦ **Guards & Middleware:** Role-based guards protect specific teacher/admin routes, returning HTTP 403 Forbidden on unauthorized breaches.

PROJECT SUMMARY

Quizminia establishes that combining server-side MVC architecture with cloud-based Large Language Models makes it possible to generate highly personalized learning spaces.

The platform scales efficiently, keeps student engagement elevated, and prevents the traditional testing dropouts seen in linear examination portals.

FUTURE ROAD MAP

- ✦ **Multi-lingual Engine:** Dynamically generate assessments in multiple regional languages.
- ✦ **AI Chat Tutor:** Interactive chat interface to discuss weaknesses immediately after quiz completion.
- ✦ **Mobile Applications:** Deploy cross-platform React Native clients.